

2.1 MATRIX OPERATIONS

Sums and Scalar Multiples

EXAMPLE 1 Let

$$A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix}$$

$A_{n \times m}$

Then

2×3

$$A + B = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix}$$

$B_{n \times m}$

$$A - B = \begin{bmatrix} 3 & -1 & 4 \\ -4 & -2 & -5 \end{bmatrix}$$

$$2B = 2 \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix}$$

$$O_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$O_{3 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

THEOREM 1

Let A , B , and C be matrices of the same size, and let r and s be scalars.

a. $A + B = B + A$ *comm.*

d. $r(A + B) = rA + rB$ *dist.*

b. $(A + B) + C = A + (B + C)$ *ass.*

e. $(r + s)A = rA + sA$

c. $A + O = A$

f. $r(sA) = (rs)A$

$$I_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Identity

Matrix Multiplication

$$AB = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & 13 & \square \end{bmatrix}$$

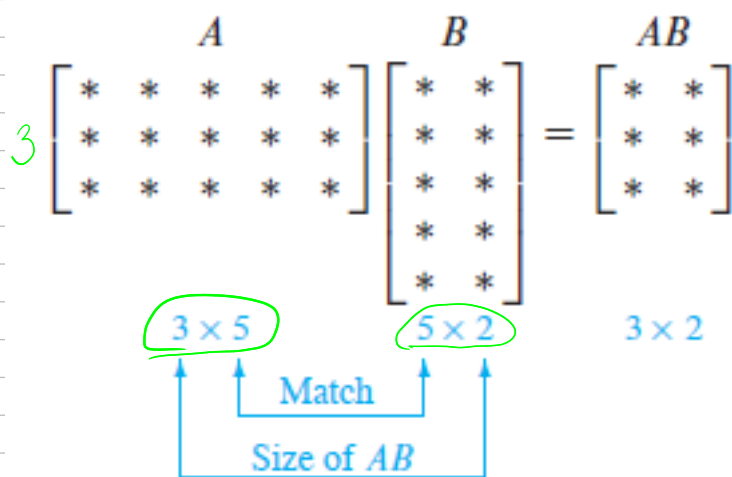
For the entry in row 2 and column 2 of AB , use row 2 of A and column 2 of B :

$$\begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & 1(3) + -5(-2) & \square \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & 13 & \square \end{bmatrix}$$

ROW-COLUMN RULE FOR COMPUTING AB

If the product AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B . If $(AB)_{ij}$ denotes the (i, j) -entry in AB , and if A is an $m \times n$ matrix, then

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$



$$A_{n \times m} \cdot B_{m \times k} = C_{n \times k}$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 2 & 5 \end{bmatrix} \stackrel{?}{=} \text{?}$$

$$2 \times 2 \cdot 3 \times 2$$

THEOREM 2

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.

- $A(BC) = (AB)C$ (associative law of multiplication)
- $A(B + C) = AB + AC$ (left distributive law)
- $(B + C)A = BA + CA$ (right distributive law)
- $r(AB) = (rA)B = A(rB)$ for any scalar r
- $I_m A = A = A I_n$ (identity for matrix multiplication)

WARNINGS:

- In general, $AB \neq BA$.
- The cancellation laws do *not* hold for matrix multiplication. That is, if $AB = AC$, then it is *not* true in general that $B = C$. (See Exercise 10.)
- If a product AB is the zero matrix, you *cannot* conclude in general that either $A = 0$ or $B = 0$. (See Exercise 12.)

$$\begin{bmatrix} 1 & 2 \\ 0 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 1 & 1 \cdot (-1) + 2 \cdot 0 \\ 0 \cdot 1 + 5 \cdot 1 & 0 \cdot (-1) + 5 \cdot 0 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 1 & 2 \end{bmatrix} \neq \begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix}$$

For numbers (real) to

For matrices

$$3 \cdot 5 = 3 \cdot x \Rightarrow x = 5 \text{ ok}$$

$$A \cdot B = A \cdot C \text{ (cannot cancel } A) \neq B = C$$

The Transpose of a Matrix

Given an $m \times n$ matrix A , the **transpose** of A is the $n \times m$ matrix, denoted by A^T , whose columns are formed from the corresponding rows of A .

$$A_{n \times m} \Rightarrow A^T_{m \times n}$$

EXAMPLE 8 Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -3 & 5 & -2 & 7 \end{bmatrix}$$

Then

$$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}, \quad B^T = \begin{bmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 & -3 \\ 1 & 5 \\ 1 & -2 \\ 1 & 7 \end{bmatrix}$$

Let A and B denote matrices whose sizes are appropriate for the following sums and products.

- $(A^T)^T = A$ ✓
- $(A + B)^T = A^T + B^T$ ✓
- For any scalar r , $(rA)^T = rA^T$
- $(AB)^T = B^T A^T$

2.2 THE INVERSE OF A MATRIX

An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C such that

$$\underline{CA} = I \quad \text{and} \quad AC = I \quad AC = CA = I$$

where $I = I_n$, the $n \times n$ identity matrix. In this case, C is an **inverse** of A . In fact, C is uniquely determined by A , because if B were another inverse of A , then $B = BI = B(AC) = (BA)C = IC = C$. This unique inverse is denoted by A^{-1} , so that

$$\underline{A^{-1}A} = I \quad \text{and} \quad AA^{-1} = I$$

A matrix that is *not* invertible is sometimes called a **singular matrix**, and an invertible matrix is called a **nonsingular matrix**.

EXAMPLE 1 If $A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$ and $C = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}$, then

$$\underline{AC} = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and}$$

$$\underline{CA} = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus $\underline{C = A^{-1}}$.

THEOREM 6

- a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A \quad \checkmark$$

- b. If A and B are $n \times n$ invertible matrices, then so is AB , and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}$$

- c. If A is an invertible matrix, then so is A^T , and the inverse of A^T is the transpose of A^{-1} . That is,

$$(A^T)^{-1} = (A^{-1})^T \quad \checkmark$$

An Algorithm for Finding A^{-1}

If we place A and I side by side to form an augmented matrix $[A \ I]$, then row operations on this matrix produce identical operations on A and on I . By Theorem 7, either there are row operations that transform A to I_n and I_n to A^{-1} or else A is not invertible.

ALGORITHM FOR FINDING A^{-1}

Row reduce the augmented matrix $[A \ I]$. If A is row equivalent to I , then $[A \ I]$ is row equivalent to $[I \ A^{-1}]$. Otherwise, A does not have an inverse.

EXAMPLE 7 Find the inverse of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$, if it exists.

Inter
swapping
Replac

$$\begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{bmatrix} \xrightarrow{R_3 - 4R_1, R_3 + 3R_2}$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{bmatrix} \xrightarrow{R_1 - 3R_3, R_2 - R_3, R_3/2} \begin{bmatrix} 1 & 0 & 0 & -4.5 & 7 & -1.5 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 1.5 & -2 & 0.5 \end{bmatrix}$$

$$\begin{aligned} -6 + 3 + 4 \\ -18 + 6 + 12 \\ 28 - 12 - 16 \end{aligned}$$

Check $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \cdot \begin{bmatrix} -4.5 & 7 & -1.5 \\ -2 & 4 & -1 \\ 1.5 & -2 & 0.5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ✓

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \cdot \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} = I \quad \text{check!!}$$

THEOREM 8

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

- a. A is an invertible matrix. → 3 row oper
- b. A is row equivalent to the $n \times n$ identity matrix. ↖
- c. A has n pivot positions. ↖
- d. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- e. The columns of A form a linearly independent set.
- f. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
- g. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
- h. The columns of A span $\mathbb{R}^n$.
- i. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- j. There is an $n \times n$ matrix C such that $CA = I$.
- k. There is an $n \times n$ matrix D such that $AD = I$.
- l. A^T is an invertible matrix.

2.5 MATRIX FACTORIZATIONS

$$x^2 + 2xy + y^2: 1+1+2+1+1=6$$

$$(x+y)^2: 1+1=2$$

The LU Factorization

The LU factorization, described below, is motivated by the fairly common industrial and business problem of solving a sequence of equations, all with the same coefficient matrix:

$$Ax = b_1, \quad Ax = b_2, \quad \dots, \quad Ax = b_p \quad (1)$$

See Exercise 32, for example. Also see Section 5.8, where the inverse power method is used to estimate eigenvalues of a matrix by solving equations like those in sequence (1), one at a time.

When A is invertible, one could compute A^{-1} and then compute $A^{-1}b_1, A^{-1}b_2,$ and so on. However, it is more efficient to solve the first equation in sequence (1) by row reduction and obtain an LU factorization of A at the same time. Thereafter, the remaining equations in sequence (1) are solved with the LU factorization.

At first, assume that A is an $m \times n$ matrix that can be row reduced to echelon form, *without row interchanges*. (Later, we will treat the general case.) Then A can be written in the form $A = LU$, where L is an $m \times m$ lower triangular matrix with 1's on the diagonal and U is an $m \times n$ echelon form of A . For instance, see Figure 1. Such a factorization is called an **LU factorization** of A . The matrix L is invertible and is called a *unit lower triangular matrix*.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix} \begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Unit L over U upper

FIGURE 1 An LU factorization.

Before studying how to construct L and U , we should look at why they are so useful. When $A = LU$, the equation $Ax = b$ can be written as $L(Ux) = b$. Writing y for Ux , we can find x by solving the pair of equations

$$\begin{cases} Ly = b \\ Ux = y \end{cases} \quad (2)$$

First solve $Ly = b$ for y , and then solve $Ux = y$ for x . See Figure 2. Each equation is easy to solve because L and U are triangular.

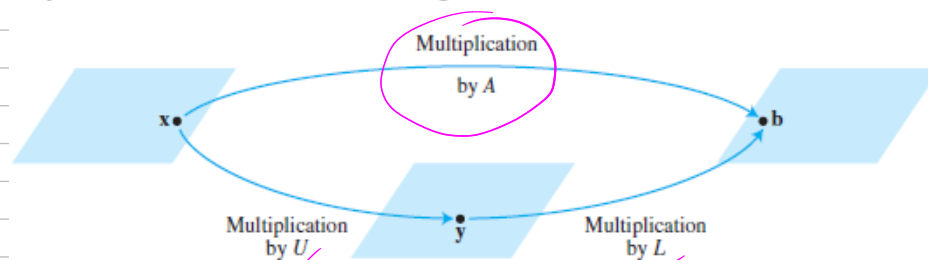


FIGURE 2 Factorization of the mapping $x \mapsto Ax$.

ALGORITHM FOR AN LU FACTORIZATION

1. Reduce A to an echelon form U by a sequence of row replacement operations, if possible.
2. Place entries in L such that the same sequence of row operations reduces L to I .

EXAMPLE 2 Find an LU factorization of

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -3 & -2 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 2 & -4 & 2 \\ 1 & 5 & -4 \\ -6 & -2 & 4 \end{bmatrix} = L \cdot U$$

$$\begin{bmatrix} 2 & -4 & 2 \\ 1 & 5 & -4 \\ -6 & -2 & 4 \end{bmatrix} \begin{array}{l} \\ R_2 - \frac{1}{2}R_1 \\ R_3 - (-3)R_1 \end{array} \quad \begin{bmatrix} 2 & -4 & 2 \\ 0 & 7 & -5 \\ 0 & -19 & 10 \end{bmatrix} \begin{array}{l} \\ \\ R_3 - (-2)R_2 \end{array}$$

$$\begin{bmatrix} 2 & -4 & 2 \\ 0 & 7 & -5 \\ 0 & 0 & 0 \end{bmatrix} \text{ Upper } \Rightarrow \begin{array}{l} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -3 & -2 & 1 \end{bmatrix} \begin{array}{l} \downarrow \downarrow \downarrow \\ \begin{bmatrix} 2 & -4 & 2 \\ 0 & 7 & -5 \\ 0 & 0 & 0 \end{bmatrix} \end{array} \\ \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -3 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & -4 & 2 \\ 0 & 7 & -5 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -4 & 2 \\ 1 & 5 & -4 \\ -6 & -2 & 4 \end{bmatrix} A \end{array}$$